

04 — GPs for Navigation and Sensor Fusion

What you'll learn: how dead-reckoning and the error-state EKF work in brief, the concrete ways a GP improves DVL/IMU fusion, and a walkthrough of the published underwater-GP results and what transfers to our sensor setup.

Prereqs: 01-03.

Why it matters here: this is the bridge from "what is a GP" to "what we actually build."

1. The underwater navigation problem, briefly

With no GPS, an AUV estimates pose by **dead-reckoning**: integrate IMU specific force/angular rate

(at ~100 Hz) for a high-rate but **drifting** estimate, and correct it with **DVL** body-frame velocity (~1 Hz) and a **magnetometer** heading. This repo does exactly that with an error-state

(indirect) Extended Kalman Filter in `dvl_correction/src/dvl_imu_kalman.py`.

Error-state EKF in one paragraph

Rather than filtering the full nonlinear state, the error-state EKF tracks a small **error** between the true state and the IMU-propagated nominal state. IMU samples **propagate** the nominal state and

grow the error covariance; DVL/mag measurements **update** (shrink) it. Our filter's 16-element error

state is `[position(3), velocity(3), attitude(3), gyro-bias(3), accel-bias(3), declination(1)]`. Each

measurement update needs three things: a **residual** (measurement – prediction), a measurement model

H, and a **measurement-noise covariance R**. *The GP's job is to supply better DVL velocity values and, crucially, a better, time-varying R.* (Code anchors in `implementation/01.`)

2. Four ways a GP improves DVL/IMU fusion

1. **Denoising.** Replace each raw DVL velocity with the GP posterior **mean** over a recent window.

The GP smooths sensor noise using temporal (and cross-axis) correlation, so the EKF ingests a cleaner measurement.

2. **Adaptive measurement noise R_k .** Feed the GP's predictive **covariance** as the EKF's R for

that update — per measurement, instead of a fixed constant. When the GP is confident, R is small and the filter leans on the DVL; when not (sparse/odd data), R grows and the filter protects itself. **This is the single highest-value idea** and the core of the published gains.

3. **Outlier rejection.** Score each incoming DVL sample by its **Mahalanobis distance** to the GP's

predictive band; reject the gross ones before they corrupt the filter. (Our EKF already has a Mahalanobis gate; the GP adds a second, smarter pre-filter.)

4. Dropout bridging.

During DVL silence, **predict** velocity from the GP at the expected DVL cadence, with covariance that **grows with the gap length**. The filter keeps getting velocity updates but trusts them less and less — a smooth fallback to IMU-only, rather than a hard switch.

All four use only the GP's two outputs (mean, covariance) from doc 03.

3. Why "predicted covariance as R_k " is principled, not a hack

A Kalman update is optimal **only if R reflects the true measurement uncertainty**. A fixed R is wrong whenever conditions change (which underwater they always do). The GP's predictive variance is a calibrated estimate of how uncertain this particular velocity value is, given the recent data and learned dynamics. Plugging it in makes the filter's uncertainty bookkeeping honest — exactly what the error-state EKF assumes. This is why the approach improves not just velocity but downstream states like orientation: better-weighted velocity updates propagate into a better attitude estimate.

4. What the literature reports

- Cohen & Klein, GPR for Improved Underwater Navigation (2025). **A** multi-output GP regression (MOGPR) **estimates DVL velocity and its covariance, fed into an error-state EKF as velocity measurement + adaptive R_k (loosely-coupled)**. **Reported** $\approx$ 20% velocity-RMSE reduction and up to **42% orientation** improvement on field trajectories. Kernel: a sum of ARD squared-exponential
 - Matérn-3/2 + rational-quadratic, trained by gradient descent (Adam). Honest caveat: MOGPR stores the full training set and inverts a large matrix — **expensive in real time** (motivates our windowing + edge plan).
 - **GP-UKF (Ko & Fox lineage)**. Use a GP to learn the motion/observation model and run it inside an Unscented Kalman Filter. Better tracking and graceful degradation than a fixed parametric model when dynamics are hard to model analytically.
 - **GP for outlier-robust filtering**. GPR regresses "pseudo-measurements" to replace detected outliers in changeable marine environments (robustness under disturbance).

Full citations: doc 09.

5. What transfers to **our** setup (and one key difference)

Aspect	Cohen & Klein 2025	This repo
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GP <i>inputs</i>	raw <i>4-beam</i> radial velocities	<i>already-resolved 3-D velocity</i> (beams are only validity flags)
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GP outputs	3-D DVL velocity + covariance	same (3-D body-frame velocity + 3x3 covariance)
Fusion	error-state EKF, adaptive R_k	identical pattern; our EKF already accepts a per-measurement <code>covariance_body</code>
Kernel	ARD-SE + Matérn-3/2 + RQ	same composite, made multitask for cross-axis correlation

The one real difference: **our DVL exposes resolved velocity, not raw beams**. So our GP is a

temporal model time → velocity (denoise/bridge/outlier), rather than a beam→velocity map. This

is simpler and still delivers denoising, adaptive R, outlier rejection, and—uniquely—gap bridging, which the resolved-velocity temporal form is especially good at. If raw beams are exposed

later, the beam→velocity MOGPR can be added as an input-space variant.

6. Loosely- vs tightly-coupled (our choice)

- **Loosely coupled** (our plan): the GP produces a velocity measurement + R; the EKF treats it as

it already treats DVL. Minimal, low-risk, **no EKF-core changes**.

- **Tightly coupled**: fold GP states into the filter (e.g. GP latent force). More powerful, much more

invasive. Out of scope for the first build.

7. Failure modes to respect

- **Garbage in, garbage out**: if the window is full of outliers, the GP learns them. Mitigate with

the EKF gate + GP outlier pre-filter + robust hyperparameter bounds.

- **Over-smoothing**: too-large lengthscale erases real manoeuvres. The learned lengthscale and the

Matérn-3/2 choice guard against this; validate on the benchmark.

- **Over-confident bridging**: if the predictive variance grows too slowly, the filter over-trusts a

long-bridged value. Cap bridging duration (`max_bridge_s`) and verify variance growth on the benchmark (`implementation/03`).

Key takeaways

- A GP improves DVL/IMU fusion four ways: **denoise, adaptive R_k , outlier rejection, gap bridging**.

- **Adaptive R_k** (predicted covariance as measurement noise) is the principled core and the source

of published gains; it even improves orientation downstream.

- Published results (~20–42% gains) use the same EKF + adaptive-R pattern we plan; the only

difference is we model **resolved** velocity over time (our DVL gives no raw beams), which is simpler and excels at **bridging**.

- We use the **loosely-coupled** form: **no EKF-core changes**.

Further reading

- Cohen & Klein 2025 (arXiv:2502.16510); Ko & Fox, GP-BayesFilters; see doc 09.

- Next: 05 — State-space GPs and the Kalman duality.